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Six Categories of Research Experience

- Scale Development
- Psychometrics Work
- Latent Growth Models/SEM Trees
- Program Evaluation
- Big-Fish-Little-Pond Effect Research
- Additional Studies with Large-Scale Assessments

Research Experience Category 1 - Scale Development

- The Classroom Engagement Inventory[1]
- Science Communication Training Effectiveness[2]
- Multidimensional Life Satisfaction of Internal Migrant Workers' Children in China[3]

[1]**Wang, Z.**, Bergin, C., & Bergin, D. A. (2014). Measuring engagement in fourth to twelfth grade classrooms: The Classroom Engagement Inventory. *School Psychology Quarterly, 29(4), 517-535. https://doi.org/10.1037/spq0000050*

[2]Rodgers, S., **Wang, Z.**, & Schultz, J. C. (2020). A scale to measure science communication training effectiveness. *Science Communication, 42(1), 90-111.* https://doi.org/10.1177/1075547020903057

[3](Dissertation Advisor: Ze Wang) Liu, J. (2022). *Development and preliminary validation of multidimensional life satisfaction of internal migrant workers' children in China* (Publication No. 29398822) [Doctoral dissertation, University of Missouri-Columbia]. ProQuest Dissertations Publishing.

Research Experience Category 2 - Psychometrics

- Positive and Negative Affect Schedule (PANAS)[1] – CFA models with Bayesian Methods
- Wechsler Individual Achievement Test, 3rd Edition (WIAT-III)[2] – CFA models with a special population
- Career Futures Inventory-Revised (CFI-R)[3] – Measurement Invariance
- Self-Description Questionnaire (SDQ)[4] – Bifactor model with extended dimensions
- Woodcock–Johnson Psychoeducational Battery–Third Edition (WJ III)[5] – Measurement Invariance
- Teacher Observation of Classroom Adaptation Checklist (TOCA-C)[6] – Comparison of Five Measurement Models; Measurement Invariance

- [1]Kim, M., & **Wang, Z.** (2022). Factor structure of the PANAS with Bayesian structural equation modeling in a Chinese sample. *Evaluation and The Health Professions*.
- [2]Parkin, J. R., & **Wang, Z.** (2021). Confirmatory factor analysis of the WIAT-III in a referral sample. *Psychology in the Schools*, *58*(5), 837-852. https://doi.org/https://doi.org/10.1002/pits.22474
- [3]Park, C. J., Rottinghaus, P. J., **Wang, Z.**, Falk, N., Zhang, T., & Ko, S.-J. (2019). Measurement invariance of the Career Futures Inventory-Revised across general and client samples. *Journal of Career Assessment, 27*(4), 711-725. doi: https://doi.org/10.1177/1069072718816514
- [4]**Wang, Z.**, & Guo, Y. Y. (2017). Deriving a structural model for the multidimensional self-concept construct: A case of middle school students in mainland China. In M. Williams (Ed.), *Self-concept: Perceptions, cultural influences and gender differences* (pp. 77-106). Hauppauge, NY: Nova Publishers.
- [5]Frisby, C. L., & **Wang, Z.** (2016). The g factor and cognitive test session behavior: Using a latent variable approach in examining measurement invariance across age groups on the WJ III. *Journal of Psychoeducational Assessment, 34*(6), 524-535. doi: 10.1177/0734282915621440
- [6]**Wang, Z.**, Rohrer, D., Chuang, C.-C., Fujiki, M., Herman, K., & Reinke, W. (2015). Five methods to score the teacher observation of classroom adaptation checklist and to examine group differences. *The Journal of Experimental Education, 83*(1), 24-50. doi: 10.1080/00220973.2013.876230

Research Experience Category 3 - Latent Growth Models/SEM Trees

- **Wang, Z.** (2022). A comparison of three approaches to covariate effects on latent factors. *Large-scale Assessments in Education, 10*(26). https://doi.org/10.1186/s40536-022-00148-2

- **Wang, Z.** & Su, I. (2013). Longitudinal factor structure of general self-concept and locus of control among high school students. *Journal of Psychoeducational Assessment, 31*(6), 554-565. doi: 10.1177/0734282913481651

- Frisby, C., & **Wang, Z.** (2018). An evaluation of the word triad method for monitoring spelling progress. *Journal of Applied School Psychology, 34*(2), 101-133. doi: 10.1080/15377903.2017.1381660

Research Experience Category 4 - Program Evaluation

- **Wang, Z.**, Zhang, T., Liu, J., & Yonke, S. (2019). Co-teaching Chinese in middle schools and high schools. *Chinese as a Second Language*, *54*(1), 1-30. Doi: https://doi.org/10.1075/csl.17027.wan

- Rodgers, S, **Wang, Z.**, Maras, M., Burgoyne, S., Balakrishnan, B., Stemmle, J., & Schultz, J. C. (2018). Decoding Science: Development and evaluation of a science communication training program using a triangulated framework. *Science Communication, 40(1)*, 3-32. doi: 10.1177/1075547017747285

- **Wang, Z.**, Tsai, C. L., & McFarling, P. (2017). *Evaluation of University of Missouri's Instruction and Course Evaluation*. Columbia, MO: University of Missouri, Assessment Resource Center.

- **Wang, Z.** & Hampton, N. J. (2017). *Missouri prediabetes media campaign evaluation*. Columbia, MO: University of Missouri, Health Communication Research Center.

Research Experience Category 5 - Big-Fish-Little-Pond Effect Research with TIMSS Data

- **Wang, Z.** (2020). When large-scale assessments meet data science: The big-fish-little-pond effect in fourth- and eighth-grade mathematics across nations. *Frontiers in Psychology*, *11*(2427). https://doi.org/10.3389/fpsyg.2020.579545

- **Wang, Z.**, & Bergin, D. (2017). Perceived relative standing and the big-fish-little-pond effect in 59 countries and regions: Analysis of TIMSS 2011 data. *Learning and Individual Differences, 57*, 141-156. doi: 10.1016/j.lindif.2017.04.003

- **Wang, Z.** (2015). Examining big-fish-little-pond-effects across 49 countries: a multilevel latent variable modelling approach. *Educational Psychology, 35*(2), 228-251. doi: 10.1080/01443410.2013.827155

Research Experience Category 6 - Additional Studies with Large-Scale Assessments

- Trends in International Mathematics and Science Study (TIMSS)[1]
- National Educational Longitudinal Study of 1988 (NELS:88)[2]
- Progress in International Reading Literacy Study (PIRLS)[3]
- National Longitudinal Study of Adolescent Health (Add Health)[4]

[1]**Wang, Z.**, Osterlind, S., & Bergin, D. (2012). Building mathematics achievement models in four countries using TIMSS 2003. *International Journal of Science and Mathematics Education, 10*(5), 1215-1242. doi: 10.1007/s10763-011-9328-6

[2]**Wang, Z.** & Su, I. (2013). Longitudinal factor structure of general self-concept and locus of control among high school students*. Journal of Psychoeducational Assessment, 31*(6), 554-565. doi: 10.1177/0734282913481651

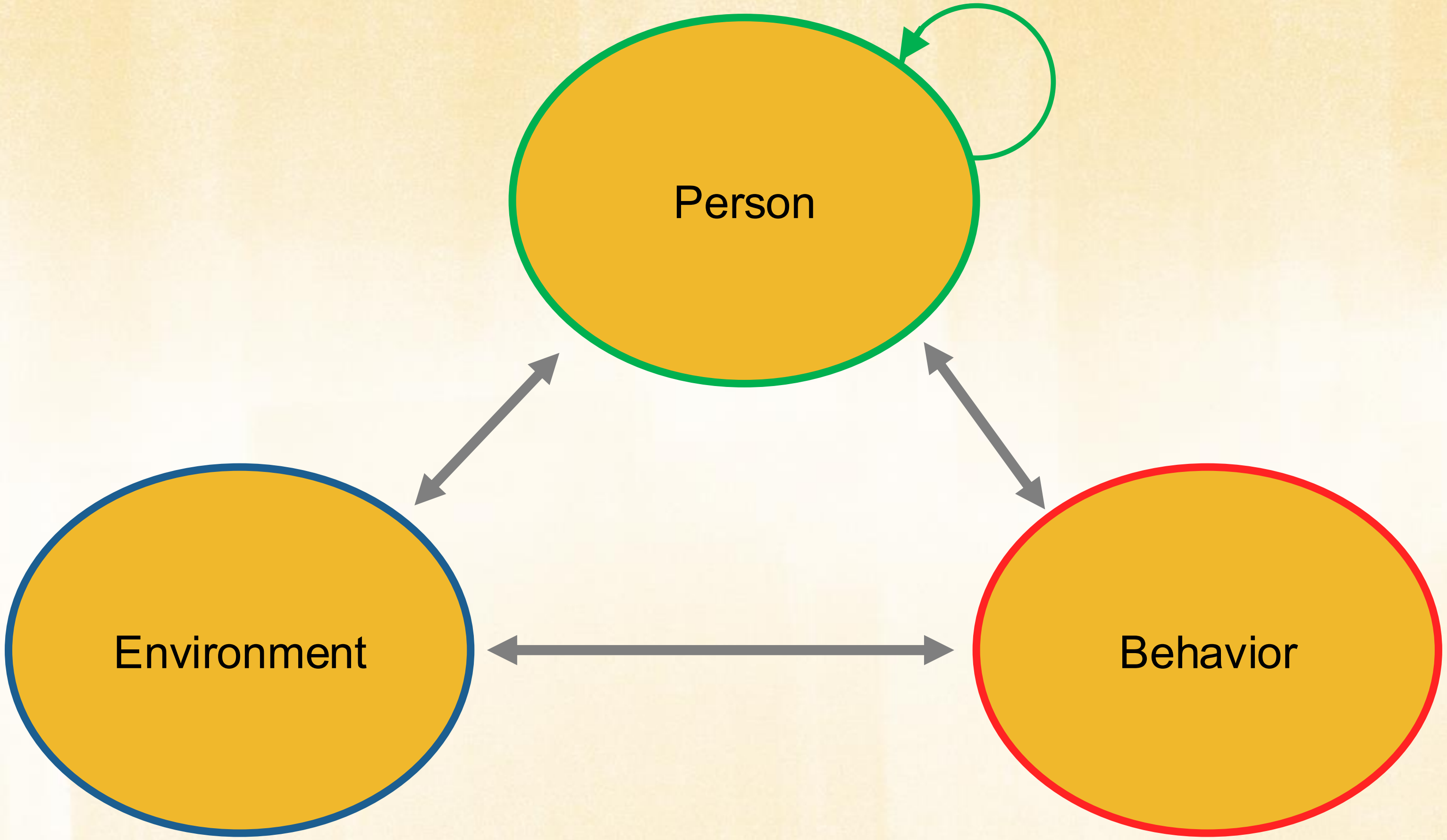
[3]Wang, Q., **Wang, Z.**, & Osterlind, S. J. (2011). Modeling the effects of home environment, reading self-concept and attitude on text comprehension between British and Chinese students. *New Waves – Educational Research & Development, 14(1),* 22-33.

[4]Whitney, S. D., Prewett, S., **Wang, Z.**, & Chen, H. (2017). Father's importance in adolescents' academic achievement. *International Journal of Child, Youth and Family Studies, 8*(3-4), 101-126. doi: http://dx.doi.org/10.18357/ijcyfs83/4201718073

Mendoza, P., Zhou, E., Abdelmalek, N., & **Wang, Z.** (2017). Higher education surveys from United States' National Center for Education Statistics. *International Journal of Quantitative Research in Education, 4*(1/2), 3-30. doi: 10.1504/IJQRE.2017.10007531

Guiding Theories and Theoretical Framework

Social Cognitive Theory



Self-Determination Theory

Self-Determination Theory

Cognitive Evaluation Theory

Predicts the effect that any external event will have on intrinsic motivation. Explains why some external events support autonomy, competence, and intrinsic motivation, while other events interfere with and thwart these motivations.

Organismic Integration Theory

Introduces types of extrinsic motivation. Specifies the antecedents, characteristics, and consequences of each type. Explains students' successful versus unsuccessful academic socialization.

Causality Orientations Theory

Identifies individual differences in how students motivate and engage themselves. To engage themselves, some students rely on autonomous guides to action while others rely on controlling and environmental guides.

Basic Psychological Needs Theory

Elaborates the concept of psychological needs and specifies their relation to intrinsic motivation, high-quality engagement, effective functioning, and psychological well-being.

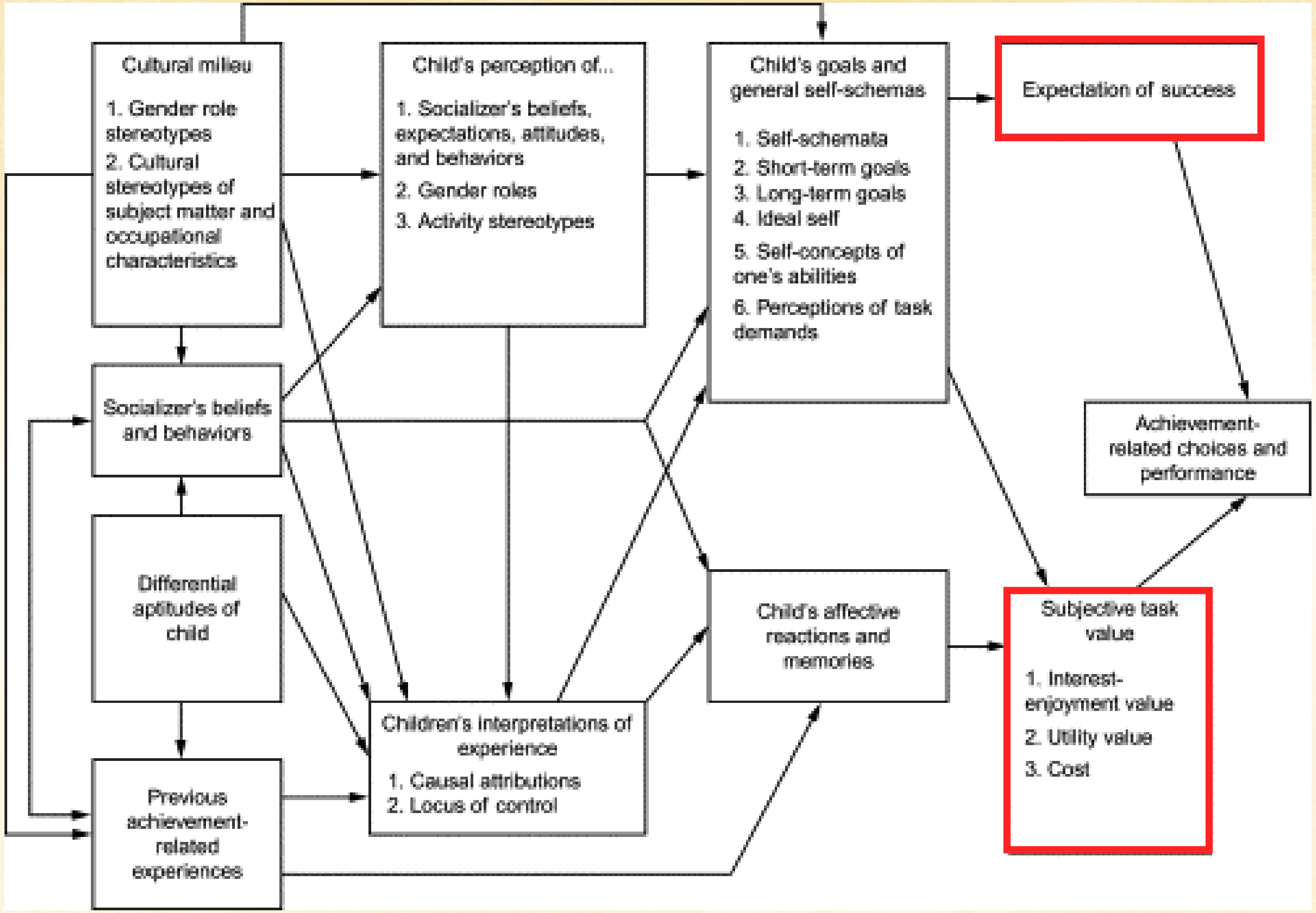
Goal Contents Theory

Distinguishes intrinsic goals from extrinsic goals to explain how the former supports psychological needs and well-being whereas the latter neglects these needs and fosters ill being.

Relationships Motivation Theory

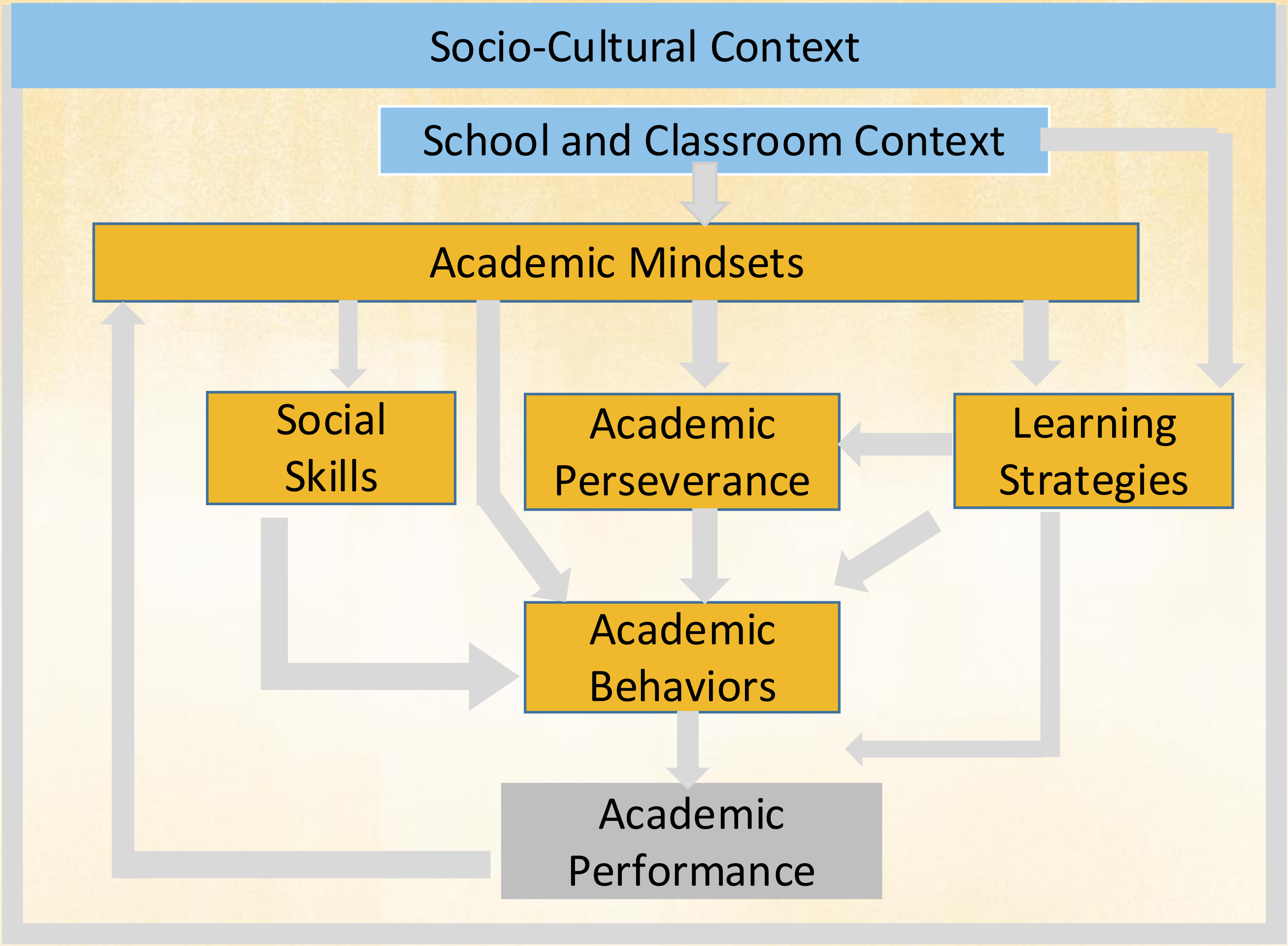
Explains the development and maintenance of close personal relationships.

Expectancy-Value Model



Source: Petersen, J., & Hyde, J. S. (2014). Chapter Two - Gender-Related Academic and Occupational Interests and Goals. In L. S. Liben & R. S. Bigler (Eds.), *Advances in Child Development and Behavior (Vol. 47, pp. 43-76). JAI. https://doi.org/https://doi.org/10.1016/bs.acdb.2014.04.004*

Framework of Noncognitive Factors for Academic Performance



Adapted from Farrington, C. A., Roderick, M., Allensworth, E., Nagaoka, J., Keyes, T. S., Johnson, D. W., & Beechum, N. O. (2012). *Teaching adolescents to become learners. The role of noncognitive factors in shaping school performance: A critical literature review. University of Chicago Consortium on Chicago School Research. http://ccsr.uchicago.edu/publications/teaching-adolescents-become-learners-role-noncognitivefactors-shaping-school*